

vLive User Guide

Professional Live Video Processing Platform

Document Information

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Table 1: Document Metadata

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Platform Overview

vLive is a comprehensive live video processing platform designed to streamline the end-to-end live streaming workflow. It supports multicast and unicast ingest, adaptive bitrate transcoding, AES, DASH and HLS packaging, DRM protection, origin management, CDN integration, and real-time monitoring.

Core Platform Capabilities

- Multicast and unicast ingest support for diverse input sources
- Adaptive Bitrate (ABR) transcoding with multiple profiles
- Multi-format packaging (AES, DASH, HLS)
- DRM protection with FairPlay and Widevine support
- Origin server management for content distribution
- CDN integration and media delivery optimization
- Real-time system monitoring and status tracking

Key Benefits

1. Efficient channel management across multiple formats
 2. Maintained service uptime and reliability
 3. High-quality content delivery across devices
 4. Comprehensive workflow automation
 5. Real-time system visibility and control
 6. Multi-device support (mobile, web, connected TV)
 7. Scalable architecture for growing demands
-

Dashboard Overview

The Dashboard provides a comprehensive, real-time view of your live streaming system performance. It enables quick identification of issues and facilitates rapid response to operational challenges.

Dashboard Status Indicators

The Dashboard displays the status of each workflow component using standardized color indicators:

Status Indicator	Color	Meaning
OK	Green	Component operating normally
Attention	Yellow	Requires monitoring or attention
Failure	Red	Immediate action required

Table 2: Dashboard Status Indicators and Meanings

Resource Status Overview

The Dashboard tracks comprehensive metrics for all system components:

Resource Type	Metrics Tracked	Status States
Servers	Total count, availability	Online / Offline
Transcoders	Total, utilization	Running, Stopped, Failing, Disabled
AES Packagers	Total, active streams	Running, Stopped, Failing, Disabled
DASH Packagers	Total, manifest status	Running, Stopped, Failing, Disabled
HLS Packagers	Total, segment creation	Running, Stopped, Failing, Disabled
Origins	Total, disk space	Running, Stopped, Failing, Disabled

Table 3: Resource Status Monitoring Metrics

Dashboard Use Cases

- Monitor overall system health and component status
- Identify failing or degraded components quickly
- Track resource utilization across servers
- Respond to alerts and critical events
- Maintain service level agreements (SLAs)
- Plan capacity upgrades based on trends

Input Stream Management

An Input Stream represents the entry point for each channel in vLive. It connects your live feed (e.g., UDP, RTMP) to the platform for processing and distribution. Proper input stream configuration is critical for channel operation and reliability.

Understanding Input Streams

- Each channel requires a unique input stream configuration
- Input streams connect live feeds (UDP, RTMP) to vLive platform
- Failed inputs prevent channel playback and service delivery
- Input health monitoring is essential for service continuity
- Content ID uniqueness ensures proper transcoding and packaging
- Input streams form the foundation of the entire workflow

Steps to Add a New Input Stream

1. Navigate to **Input Streams** module from the left Sidebar
2. Click **Add Input Stream** button located at top-right

3. Enter a clear, identifiable **Channel Name** (examples: Sony Max HD, Star Sports 1, Discovery Kids)
4. Enter **Input Address** with supported protocol:
 - UDP format: `udp://239.x.x.x:port` (multicast)
 - RTMP format: `rtmp://source.example.com/live` (unicast)
 - Ensure the URL is valid and network-reachable
 - Verify source is active before registration
5. Enter a unique **Content ID** (mandatory field, used by all downstream services):
 - Format: Alphanumeric identifier (example: OQTOzZkBoRDJ74Ryniuo)
 - Must be unique across all channels
 - Used by transcoders, packagers, and DRM services
 - Cannot be reused once assigned
6. Click **Save** to register the input stream

Input Stream Configuration Fields

Field	Type	Description
Channel Name	Text	Identifiable name for the live channel
Input Address	URL	Source URL (UDP or RTMP format)
Content ID	Unique ID	Identifier for transcoding and packaging
Transcoders	Link	Associated transcoder configurations
AES Packagers	Link	Associated AES packaging configurations
DASH Packagers	Link	Associated DASH packaging configurations
HLS Packagers	Link	Associated HLS packaging configurations

Table 4: Input Stream Configuration Fields Reference

Input Stream List Actions

Action	Icon	Function
View	Eye	Open Input Stream Details in read-only mode
Edit	Pencil	Modify input stream configuration
Delete	Bin	Remove input stream from platform

Table 5: Input Stream Action Options

View Input Stream Details

When viewing an input stream in detail, operators can access the following information and perform these actions:

Available Information:

- **Input Stream Details:** Name, Input Address, Content ID
- **Associated Resources:** Linked transcoders and all packager configurations
- **Configuration Status:** Identifies which components are configured and which are missing
- **Resource Health:** Status of associated transcoders and packagers

Available Actions (Top Buttons):

- **Edit** - Opens configuration page to modify stream parameters
- **Back** - Returns to Input Stream list view

Edit Input Stream

The Edit action allows modification of input stream parameters:

Field	Modifiable	Action / Impact
Channel Name	Yes	Update stream or channel identifier
Input Address	Yes	Change UDP or RTMP source URL (affects live feed)
Content ID	Yes	Update unique identifier (used by DRM and packagers)

Table 6: Editable Input Stream Fields

Editing Workflow:

1. Navigate to the input stream and click the pencil (Edit) icon
2. Modify the desired fields
3. Review changes for accuracy
4. Click **Update** to save changes immediately
5. Or click **Cancel** to discard all changes without saving

Important Considerations:

- Address changes affect live stream ingestion immediately
- Content ID changes may affect downstream components
- Changes are applied in real-time to active streams
- Monitor dashboard for any disruptions after editing

Delete

The Delete action removes the input stream from the platform.

Deleting Workflow:

1. Navigate to the input stream and click the bin (Delete) icon
 2. A confirmation message will appear
 3. Click **Ok** to remove the input stream from the platform
 4. Or click **Cancel** to discard the action
-

Transcoder Management

The Transcoder module converts raw input streams into multiple Adaptive Bitrate (ABR) renditions, enabling players to adapt playback quality to varying network conditions. Each transcoder runs on a dedicated server and produces HD/SD profiles based on the selected ABR configuration.

Transcoder Functions and Benefits

- Converts raw input into multiple ABR renditions for quality adaptation
- Produces HD and SD profiles simultaneously
- Enables network-optimized playback experience
- Supports real-time quality switching based on bandwidth
- Required for generating AES, DASH, and HLS outputs
- Distributes processing load across servers
- Maintains consistent quality across devices

Steps to Add a New Transcoder

1. Navigate to **Transcoders** module from left Sidebar
2. Click **Add Transcoder** button at top-right
3. Enter **Transcoder Name** (recognizable identifier, example: Sony Max HD)
4. **Select Server** (choose from available transcoding servers):
 - Example: dh-transcoder-01, dh-transcoder-02, etc.
 - Server must have sufficient CPU and network resources
 - Verify server is online before selection
5. **Select Network Interface** (typically eth0 unless otherwise specified):
 - Examples: vlan200 - 10.102.1.200, vlan656 - 10.102.6.56
 - Ensure interface has connectivity to input source
 - Verify interface routes to output destinations

6. **Choose Input Stream** (select the input stream this transcoder will read from):
 - Must be an existing, configured input stream
 - Verify content source is active
7. **Select ABR Profile** (choose quality tier):
 - HD - Produces high-quality renditions (1080p, 720p, 480p, 360p)
 - SD - Produces standard-quality renditions (720p, 480p, 360p)
 - Choose based on target audience and bandwidth availability
8. **Select Deployment Version** (optional):
 - Pick the transcoder engine version available on server
 - If unsure, use default/latest available version
9. **Enable Transcoder** (toggle **Enable = ON** to activate):
 - Enabled transcoders begin processing immediately
 - Disabled transcoders remain configured but inactive
10. Click **Create** to register and start the transcoder

Transcoder List Display

Once created, the transcoder appears in the Transcoder list with the following information:

Field	Information Shown
Enabled Status	ENABLED / DISABLED
Live Status	RUNNING / STOPPED
Name	Transcoder name
Input Stream	Source stream name
ABR Profile	Selected quality profile (HD / SD)
Server	Hosting server name
Version	Deployment procedure version

Table 7: Transcoder List Information Fields

Transcoder Actions

Action	Icon	Description
View	Eye	Open Transcoder Details in read-only mode
Enable/Disable	Toggle	Start or stop the transcoder
Edit	Pencil	Modify transcoder configuration

Delete	Bin	Remove transcoder from platform
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Table 8: Transcoder Action Options

View Transcoder Details

The View action opens the Transcoder Details page displaying comprehensive information:

Available Controls (Top Buttons):

- **Disable / Enable** - Toggle transcoder state
- **Edit** - Open configuration modification page
- **Back** - Return to Transcoder list

Basic Information Displayed:

- **Name** - Transcoder identifier
- **Server** - Hosting server
- **Server Interface** - Network interface in use
- **Input Stream** - Source stream name
- **ABR Profile** - Quality profile selected
- **Live Status** - Current operation state (RUNNING / STOPPED)
- **Enabled Status** - Configuration state (ENABLED / DISABLED)
- **Deployment Version** - Engine version in use

Rendition Outputs:

For each configured rendition, the system displays:

Rendition	Output Details	Connection Info
1080p	Resolution and bitrate	Host:Port for connectivity
720p	Resolution and bitrate	Host:Port for connectivity
480p	Resolution and bitrate	Host:Port for connectivity
360p	Resolution and bitrate	Host:Port for connectivity

Table 9: Transcoder Rendition Output Information

Input Stream Probe:

- Allows inspection of input stream metadata
- Fetches technical details about programs and streams
- Enables verification of source configuration
- Useful for troubleshooting input connectivity

Edit Transcoder

The Edit action opens the configuration modification page:

Basic Configuration Fields:

Field	Type	Notes
Name	Text	Transcoder identifier
Server	Dropdown	Transcoding server selection
Server Interface	Dropdown	Network interface selection
Input Stream	Dropdown	Source input stream
ABR Profile	Dropdown	Quality profile (HD/SD)
Deployment Version	Dropdown	Transcoder version
Enable Transcoder	Toggle	Activation control

Table 10: Transcoder Configuration Fields

Rendition Output Configuration:

For each rendition, operators can configure:

- **Output Host** - Destination server IP or hostname
- **Output Port** - Unique port number for stream output
 - Must be unique across all renditions
 - Must be within allowed port range
 - Cannot conflict with other services

Applying Changes:

1. Review all modifications
2. Click **Update** to apply changes
3. Or click **Cancel** to discard changes without saving
4. Monitor transcoder status after updating

Enable / Disable Control:

- Starts or stops the transcoder immediately
- Disabled transcoders stop generating output streams
- Active streams are interrupted if transcoder is disabled
- Re-enabling restarts transcoder and stream generation

Enabled / Disabled

The Toggle action restarts the Transcoder from the platform.

Deleting Workflow:

1. Navigate to the Transcoder and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the transcoder from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the Transcoder from the platform.

Deleting Workflow:

1. Navigate to the Transcoder and click the bin (Delete) icon
2. A confirmation message will appear
3. Click **Ok** to remove the transcoder from the platform
4. Or click **Cancel** to discard the action

Packager Configuration

Packagers format transcoded streams into distribution formats required by various devices and platforms. vLive supports three primary packaging formats for comprehensive device coverage.

Packager Overview

Packager Type	Format	Device Support	Encryption
AES	AES segments + playlist	IPTV / Set-top boxes	AES-128
DASH	MPEG-DASH (.mpd)	Web / Android / Smart TV	Widevine
HLS	Apple HLS (.m3u8)	iOS / Safari / OTT	FairPlay

Table 11: Packager Types, Formats, and Support Matrix

General Packager Workflow

All packagers follow a similar workflow:

1. **Select Input Source** - Choose transcoder or custom stream URL
2. **Configure Renditions** - Select video and audio tracks to include
3. **Set Timing Parameters** - Configure segment and buffer durations

4. **Enable DRM Protection** - Optional encryption configuration
 5. **Configure Deployment** - Select version and enable packager
 6. **Create Packager** - Finalize configuration and activate
-

AES Packager

AES Packager encrypts transcoded streams using AES-128 encryption for secure IPTV delivery on compatible devices like set-top boxes and VLC players. It provides lightweight protection suitable for IPTV ecosystems.

AES Packager Overview

- Encrypts streams with AES-128 algorithm
- Generates encrypted media segments and playlist files
- Supports key server integration for dynamic key delivery
- Optimized for IPTV and set-top box playback
- Low latency suitable for live TV distribution
- Compatible with legacy IPTV infrastructure

Steps to Add AES Packager

1. Navigate to **Packagers → AES Packagers → Add AES**
2. **Select Input Source** (choose one):
 - **Existing Transcoder (recommended)** - Auto-loads transcoder list, output renditions, Content ID
 - **Custom Stream URL** - Manual entry, requires manual Content ID entry
3. **Select Transcoder** (if using transcoder source):
 - Choose the transcoder generating ABR outputs
 - System automatically displays available video/audio tracks
 - Verify output streams match your requirements
4. **Select Output Stream** (choose the desired video layer):
 - Options: 1080p, 720p, 480p, 360p (depending on ABR profile)
 - Select based on target device capabilities
5. **Verify Content ID** (auto-filled from input stream):
 - Confirm Content ID matches input stream
 - This ID is used by all downstream services
6. **Enter Keyserver Endpoint** (URL of encryption key server):
 - Example: `https://keyserver.example.com/keys`
 - Must be reachable from packager server

- Verify connectivity before creating packager
7. **Configure Segment Settings:**
 - Segment Duration (default: 6 seconds) - Duration of each media segment
 - Playlist Size (default: 10 entries) - Number of segments in playlist
 - Additional Segments (default: 5) - Extra segments for buffering
 8. **Select Deployment Version** (optional):
 - Pick version if specific features required
 - Default version recommended if unsure
 9. **Enable AES Packager** (toggle **Enable = ON**):
 - Enabled packagers start processing immediately
 - Disabled packagers remain configured but inactive
 10. Click **Create** to finalize configuration

AES Packager List Display

Once created, the AES packager appears in the list with:

Field	Information
Enabled	ENABLED / DISABLED
Live Status	RUNNING / STOPPED
Content ID	Unique stream identifier
Transcoder	Associated transcoder name
Output Stream	Selected video quality (e.g., 239.19.10.100:42223)
Server	Hosting server
Version	Deployment procedure version

Table 12: AES Packager List Information

AES Packager Actions

Action	Icon	Function
View	Eye	Display packager details
Enable/Disable	Toggle	Start or stop packager
Edit	Pencil	Modify configuration
Delete	Bin	Remove packager

Table 13: AES Packager Actions

View AES Packager Details

Top Action Buttons:

- **Disable / Enable** - Toggle packager state
- **Edit** - Modify configuration
- **Back** - Return to packager list

Configuration Details Displayed:

Parameter	Description / Example
Live Status	Runtime state (e.g., RUNNING)
Enabled	Active status (e.g., ENABLED)
Input Source Type	Source type (e.g., TRANSCODER)
Transcoder	Associated transcoder name (e.g., Janata TV)
Output Stream	Selected quality (e.g., 720p)
Keyserver Endpoint	Encryption key server URL
Segment Duration	Segment duration (e.g., 6 seconds)
Playlist Size	Segments in playlist (e.g., 10 entries)
Additional Segments	Buffer segments (e.g., 5)
Status	Deployment state (e.g., PENDING / RUNNING)
Created At	Creation timestamp
Updated At	Last modification timestamp
Playlist URL	Playback URL for clients

Table 14: AES Packager Configuration Details

Edit AES Packager

The Edit page is organized into four configuration sections:

1. Input Stream Address

- Input Source - Source type and location
- Transcoder - Associated transcoder selection
- Output Stream - Selected video quality
- Content ID - Unique stream identifier

2. Playlist Configuration

- Segment Duration - Segment length in seconds

- Playlist Size - Number of segments in live playlist
- Additional Segments - Extra segments for buffering

3. Encryption Settings

- Keyserver Endpoint - URL of key distribution server

4. Deployment

- Deployment Version - Service version selection
- Enable AES Packager - Activation toggle

Applying Changes:

1. Review all modifications carefully
2. Click **Update** to apply changes
3. Or click **Cancel** to discard without saving

Enabled / Disabled

The toggle action restarts the AES packager from the platform.

Deleting Workflow:

1. Navigate to the packager then click on AES packager and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the AES packager from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the AES packager from the platform.

Deleting Workflow:

1. Navigate to the packager then click on AES packager and click the bin (Delete) icon
2. A confirmation message will appear
3. Click **Ok** to remove the AES packager from the platform
4. Or click **Cancel** to discard the action

DASH Packager

DASH (Dynamic Adaptive Streaming over HTTP) Packager generates MPEG-DASH manifests (.mpd) and media segments for web browsers, Android devices, and smart TVs. It supports multiple renditions, audio tracks, DVR functionality, and Widevine DRM protection.

DASH Packager Overview

- Generates MPEG-DASH format for web and Android streaming
- Creates .mpd manifest files for client playlist management
- Supports multiple video and audio renditions
- Enables adaptive bitrate streaming based on network conditions
- Provides DVR and time-shift capabilities
- Integrates with Widevine DRM for content protection
- Optimized for modern web browsers and mobile devices

Steps to Add a DASH Packager

1. Navigate to **Packagers → DASH Packagers → Add DASH**
2. **Select Input Source** (choose one):
 - **Existing Transcoder (recommended)** - Auto-fills Content ID and audio tracks
 - **Custom Stream URL** - Manual URL entry
3. **Select Transcoder** (choose the transcoder to package):
 - Verify transcoder is running and producing output
 - System displays available video/audio tracks
4. **Verify/Modify Content ID** (auto-generated but editable):
 - Used by all downstream components
 - Must be unique across platform
5. **Select Video Streams** (choose renditions to include):
 - **For HD Profile:** 1080p, 720p, 480p, 360p
 - **For SD Profile:** 720p, 480p, 360p
 - Minimum recommended: 3 renditions for good UX
6. **Select Audio Streams** (choose 1 or more audio tracks):
 - Single track minimum for mono audio
 - Multiple tracks for multi-language support
7. **Configure Segment Settings:**
 - **Segment Duration** (default: 6 seconds) - Duration of each media segment
 - **Fragment Duration** (default: 6 seconds) - Internal fragment timing
 - **Min Buffer Time** (default: 3 seconds) - Minimum playback buffer
 - **Suggested Delay** (default: 12 seconds) - Playback latency
8. **Enable DVR Settings** (for time-shift capability):
 - **Time Shift Buffer** (default: 900 seconds / 15 minutes)
 - Enables rewind and pause capabilities

- Duration determines how far back users can rewind
9. **Configure DRM / Encryption:**
 - Enable DRM Packager - Toggle for encryption
 - **Required for Widevine DRM protection**
 - Select DRM provider if premium content
 10. **Select Deployment Version** (optional):
 - Choose specific version if needed
 - Default recommended if unsure
 11. **Enable DASH Packager** (toggle **Enable = ON**):
 - Activates packager for stream processing
 12. Click **Create** to finalize configuration

DASH Packager List Display

Once created, the DASH packager appears with:

Field	Information
Enabled	ENABLED / DISABLED
Live Status	RUNNING / STOPPED
Content ID	Stream identifier
Transcoder	Associated transcoder
Video Rendition	Rendition count
Audio Rendition	Rendition count
DRM Enabled	Protection status
Server	Hosting server
Version	Deployment version

Table 15: DASH Packager List Information

DASH Packager Actions

Action	Icon	Function
View	Eye	Display packager details
Disable/Enable	Toggle	Start or stop packager
Rotate Keys	Key	Rotate DRM keys without stopping stream
Edit	Pencil	Modify configuration

Delete	Bin	Remove packager
Back	Arrow	Return to list

Table 16: DASH Packager Actions

View DASH Packager Details

Top Action Buttons:

- **Disable / Enable** - Toggle packager state
- **Rotate Keys** - Rotate DRM encryption keys without stopping stream (security compliance)
- **Edit** - Modify configuration parameters
- **Back** - Return to packager list

Configuration Details Displayed:

Parameter	Description / Example
Live Status	Runtime state (e.g., RUNNING)
Enabled	Status (e.g., ENABLED)
Content ID	Stream identifier (e.g., eATozZkBoRDJ74Ryniuo)
Input Source Type	Source type (e.g., TRANSCODER)
Transcoder	Associated transcoder (e.g., Janata TV)
Video Renditions	Number of video profiles (e.g., 3 VIDEO)
Audio Renditions	Audio track count (e.g., 1 AUDIO)
Segment Duration	Duration per segment (e.g., 6 seconds)
Fragment Duration	Fragment timing (e.g., 6 seconds)
Minimum Buffer Time	Playback buffer (e.g., 3 seconds)
Suggested Delay	Playback latency (e.g., 12 seconds)
Time Shift Buffer	DVR duration (e.g., 900 seconds)
DRM Enabled	Protection status (YES / NO)
DRM Provider	DRM service (e.g., Mobiotics Prod)
Deployment Version	Version used (e.g., v2.0)
Status	Deployment state (e.g., PENDING)
Created At	Creation timestamp
Updated At	Last update timestamp

Playlist URL	DASH playback URL
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Table 17: DASH Packager Configuration Details

Edit DASH Packager

The Edit page is organized into five configuration sections:

1. Input Stream Configuration

- Input Source - Source type selection
- Transcoder - Transcoder selection
- Content ID - Stream identifier
- Video Streams - Video rendition selection
- Audio Streams - Audio track selection

2. Segment / Playlist Timing Controls

- Segment Duration - Length of each segment
- Fragment Duration - Internal fragment timing
- Minimum Buffer Time - Minimum buffer before playback
- Suggested Delay - Playback latency for stability

3. DVR / Time-Shift Controls

- Time Shift Buffer - DVR rewind duration

4. DRM / Encryption Settings

- Enable DRM Protection - Activation toggle
- DRM Provider - DRM service selection

5. Deployment Configuration

- Deployment Version - Service version selection
- Enable DASH Packager - Activation toggle

Applying Changes:

1. Review modifications carefully
2. Click **Update** to apply changes
3. Click **Cancel** to discard without saving
4. Monitor packager status after updating

Enabled / Disabled

The toggle action restarts the DASH packager from the platform.

Deleting Workflow:

1. Navigate to the packager then click on DASH packager and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the DASH packager from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the DASH packager from the platform.

Deleting Workflow:

1. Navigate to the packager then click on DASH packager and click the bin (Delete) icon
 2. A confirmation message will appear
 3. Click **Ok** to remove the DASH packager from the platform
 4. Or click **Cancel** to discard the action
-

HLS Packager

HLS (HTTP Live Streaming) Packager generates Apple HLS format (.ts segments and .m3u8 playlists) for iOS, Safari, OTT platforms, and other HLS-capable devices. It supports multiple renditions, encryption, DVR functionality, and comprehensive multi-device compatibility.

HLS Packager Overview

- Generates Apple HLS format for iOS and Safari playback
- Creates .ts media segments and .m3u8 playlist files
- Supports FairPlay DRM for encrypted delivery
- Enables DVR and time-shift functionality
- Multi-device compatibility (iOS, tvOS, web browsers)
- Adaptive bitrate streaming based on bandwidth
- Optimized for OTT and streaming service delivery

Steps to Add an HLS Packager

1. Navigate to **Packagers → HLS Packagers → Add HLS**
2. **Select Input Source** (choose one):
 - **Existing Transcoder (recommended)** - Auto-fills Content ID
 - **Custom Stream URL** - Manual URL entry
3. **Select Transcoder** (choose transcoder to package):
 - System automatically displays available video/audio tracks
 - Verify transcoder is actively producing output
4. **Verify/Modify Content ID** (auto-generated but editable):

- Unique identifier for all downstream components
 - Cannot be reused across channels
5. **Select Video Streams** (choose renditions):
 - **For HD Profile:** 1080p, 720p, 480p, 360p
 - **For SD Profile:** 720p, 480p, 360p
 - Minimum 3 renditions recommended
 6. **Select Audio Streams** (choose 1 or more tracks):
 - Single track minimum for mono audio
 - Multiple tracks for multi-language support
 7. **Configure Segment Settings:**
 - **Segment Duration** (default: 6 seconds)
 - **Fragment Duration** (default: 6 seconds)
 - **Min Buffer Time** (default: 3 seconds)
 - **Suggested Delay** (default: 12 seconds)
 8. **Enable DVR Settings** (for time-shift capability):
 - **Time Shift Buffer** (default: 900 seconds / 15 minutes)
 - Select for rewind and pause functionality
 9. **Configure DRM / Encryption:**
 - Enable DRM Packager - Toggle for FairPlay protection
 - Required for premium/encrypted content
 - Select DRM provider for Apple devices
 10. **Select Deployment Version** (optional):
 - Choose specific version if needed
 - Default recommended if unsure
 11. **Enable HLS Packager** (toggle **Enable = ON**):
 - Activates packager for stream processing
 12. Click **Create** to finalize configuration

HLS Packager List Display

Once created, the HLS packager appears with:

Field	Information
Enabled	Enabled / Disabled
Live Status	RUNNING / STOPPED
Content ID	Stream identifier

Transcoder	Associated transcoder
Video Rendition	Video quality count
Audio Rendition	Audio track count
DRM Enabled	Protection status
Server	Hosting server
Version	Deployment version

Table 18: HLS Packager List Information

HLS Packager Actions

Action	Icon	Function
View	Eye	Display packager details
Disable/Enabled	Toggle	Start or stop packager
Edit	Pencil	Modify configuration
Delete	Bin	Remove packager

Table 19: HLS Packager Actions

View HLS Packager Details

Top Action Buttons:

- **Disable / Enable** - Toggle packager state
- **Rotate Keys** - Rotate DRM encryption keys without interruption
- **Edit** - Modify configuration
- **Back** - Return to packager list

Configuration Details:

Parameter	Description / Example
Live Status	Runtime state (e.g., RUNNING)
Enabled	Status (e.g., ENABLED)
Content ID	Stream identifier (e.g., 5gTOzZkBoRDJ74Ryniqo)
Input Source Type	Source type (e.g., TRANSCODER)
Transcoder	Associated transcoder (e.g., Janata TV)

Video Renditions	Number of video profiles (e.g., 3 VIDEO)
Audio Renditions	Audio track count (e.g., 1 AUDIO)
Segment Duration	Duration per segment (e.g., 6 seconds)
Fragment Duration	Fragment timing (e.g., 6 seconds)
Minimum Buffer Time	Playback buffer (e.g., 3 seconds)
Suggested Delay	Playback latency (e.g., 12 seconds)
Time Shift Buffer	DVR duration (e.g., 900 seconds)
DRM Enabled	Protection status (YES / NO)
DRM Provider	DRM service (e.g., Mobiotics Prod)
Deployment Version	Version used (e.g., v2.0)
Status	Deployment state (e.g., PENDING)
Created At	Creation timestamp
Updated At	Last update timestamp
Playlist URL	HLS playback URL

Table 20: HLS Packager Configuration Details

Edit HLS Packager

The Edit page is organized into five configuration sections identical to DASH:

1. Input Stream Configuration

- Input Source - Source type selection
- Transcoder - Transcoder selection
- Content ID - Stream identifier
- Video Streams - Video rendition selection
- Audio Streams - Audio track selection

2. Segment / Playlist Timing Controls

- Segment Duration - Length of each segment
- Fragment Duration - Internal fragment timing
- Minimum Buffer Time - Minimum buffer before playback
- Suggested Delay - Playback latency

3. DVR / Time-Shift Controls

- Time Shift Buffer - DVR rewind duration

4. DRM / Encryption Settings

- Enable DRM Protection - Activation toggle
- DRM Provider - FairPlay provider selection

5. Deployment Configuration

- Deployment Version - Service version selection
- Enable HLS Packager - Activation toggle

Applying Changes:

1. Review all modifications
2. Click **Update** to apply changes
3. Click **Cancel** to discard without saving

Enabled / Disabled

The toggle action restarts the HLS packager from the platform.

Deleting Workflow:

1. Navigate to the packager then click on HLS packager and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the HLS packager from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the HLS packager from the platform.

Deleting Workflow:

1. Navigate to the packagers then click on HLS packager and click the bin (Delete) icon
2. A confirmation message will appear
3. Click **Ok** to remove the HLS packager from the platform
4. Or click **Cancel** to discard the action

ABR Profile Management

An ABR (Adaptive Bitrate) profile defines the set of video renditions (quality levels) that the transcoder produces. Each rendition specifies resolution, bitrate, frame rate, and codec settings. Choosing the correct profile balances quality and bandwidth consumption.

ABR Profile Overview

- Defines quality ladder for adaptive streaming

- Specifies resolution, bitrate, and codec for each rendition
- vLive supports HD and SD standard profiles
- Custom profiles can be created for specific needs
- Proper selection optimizes bandwidth usage
- More renditions provide better quality adaptation
- Fewer renditions reduce processing overhead

Common ABR Profile Examples

HD Profile (4 Renditions):

1. 1080p - 5000 Kbps - 30fps
2. 720p - 2800 Kbps - 30fps
3. 480p - 1200 Kbps - 30fps
4. 360p - 800 Kbps - 30fps

SD Profile (3 Renditions):

1. 720p - 2000 Kbps
2. 480p - 1000 Kbps
3. 360p - 600 Kbps

Steps to Add a New ABR Profile

1. Navigate to **ABR Profiles** from left Sidebar
2. Click **Add Profile** button (top-right)
3. Enter **Profile Name** (unique identifier):
 - Allowed characters: A--Z, a--z, 0--9, underscore (_), hyphen (-)
 - Length: 1--50 characters
 - Example: HD_Profile_1080p
4. Add optional **Description** (max 255 characters):
 - Example: "High Definition quality ladder for premium content"
5. **Add Renditions** (click "Add Rendition" for each quality level):
 - **Name** - Quality identifier (e.g., 1080p, 720p)
 - **Width (pixels)** - Video width (e.g., 1920)
 - **Height (pixels)** - Video height (e.g., 1080)
 - **Video Bitrate (Kbps)** - Bitrate for video (e.g., 5000)
 - **Audio Bitrate (Kbps)** - Bitrate for audio (e.g., 128)
 - **Frame Rate (fps)** - Frames per second (e.g., 25 or 30)
 - **Profile** - Encoding profile (Base / Main / High)

- **Video Codec** - Compression codec (H.264 / H.265)
 - **Audio Codec** - Audio codec (AAC / AAC3)
 - **Fit Mode** - Scaling behavior (Fit / Fill / Stretch)
6. **Add Multiple Renditions** (repeat step 4 for each quality):
 - Use "Remove" button to delete unwanted renditions
 - Arrange renditions from highest to lowest quality
 7. Click **Save** to finalize profile creation

ABR Profile List Display

Once created, the profile appears in the ABR Profiles list with:

Field	Information
Name	Profile identifier
Description	Profile purpose/description
Renditions	Number of quality levels

Table 21: ABR Profile List Information

ABR Profile Actions

Action	Icon	Function
View	Eye	Display profile details
Edit	Pencil	Modify profile configuration
Delete	Bin	Remove profile

Table 22: ABR Profile Actions

View ABR Profile Details

Header Action Buttons:

- **Edit** - Modify profile configuration
- **Back** - Return to ABR Profile list

Profile Information:

- **Name** - Unique profile identifier (e.g., el-hd)
- **Renditions** - Total number of quality levels
- **Description** - Purpose or quality ladder description

Renditions Table:

Each rendition displays complete specifications:

Rendition	Video Specs	Codec Info	Scaling
1080p	1920x1080 @ 5000 Kbps	H.264, AAC 128k	Fit/Fill/Stretch
720p	1280x720 @ 2800 Kbps	H.264, AAC 128k	Fit/Fill/Stretch
480p	854x480 @ 1200 Kbps	H.264, AAC 128k	Fit/Fill/Stretch
360p	640x360 @ 800 Kbps	H.264, AAC 128k	Fit/Fill/Stretch

Table 23: ABR Rendition Specifications

This view page is **read-only** and used for verification purposes.

Edit ABR Profile

The Edit page allows modification of profile metadata and renditions:

Basic Details:

- **Name** - Unique identifier for the profile
- **Description** - Optional text describing profile usage

Rendition Configuration:

Each rendition represents one output quality with these fields:

Field	Type	Example Values
Rendition Name	Text	1080p, 720p, 480p, 360p
Width	Integer	1920, 1280, 854, 640
Height	Integer	1080, 720, 480, 360
Video Bitrate	Integer (Kbps)	5000, 2800, 1200, 800
Audio Bitrate	Integer (Kbps)	128, 96, 64, 48
Frame Rate	Decimal (fps)	25, 30, 50, 60
Profile	Dropdown	Base, Main, High
Video Codec	Dropdown	H.264, H.265
Audio Codec	Dropdown	AAC, AAC3
Fit Mode	Dropdown	Fit, Fill, Stretch

Table 24: Rendition Configuration Fields

Rendition Actions:

- **Add Rendition** - Adds new quality level
- **Remove** - Deletes selected rendition

Applying Changes:

1. Review all modifications
2. Click **Save** to apply changes
3. Click **Cancel** to discard without saving

Delete

The Delete action removes the ABR profiles from the platform.

Deleting Workflow:

1. Navigate to the ABR profiles and click the bin (Delete) icon
2. A confirmation message will appear
3. Click **Ok** to remove the ABR profile on the platform
4. Or click **Cancel** to discard the action

DRM Provider Configuration

The DRM Providers module configures Digital Rights Management services to protect encrypted streams. vLive supports FairPlay (Apple HLS) and Widevine (DASH) DRM, ensuring only authorized devices with valid licenses can play protected content.

DRM Provider Overview

- Configures DRM services for content protection
- Supports FairPlay for iOS/Safari HLS playback
- Supports Widevine for Android/web DASH playback
- Enables license-based content access control
- Provides key management and encryption
- Integration with packagers for encrypted delivery
- Essential for premium and premium content

Steps to Add a DRM Provider

1. Navigate to **DRM Providers** from left Sidebar
2. Click **Add DRM Provider** button
3. Enter **Provider Name** (unique identifier):
 - Example: Mobiotics Prod, DRM-FairPlay, DRM-Widevine
4. **Select Provider Type** (from dropdown):

- Available options depend on system configuration
 - Example: Mobiotics (supports FairPlay and Widevine)
5. Add optional **Description**:
 - Example: "DRM for production environment"
 - Identifies purpose and usage context
 6. **Enable the Provider** (toggle **Enabled = ON**):
 - Activates provider for use in packager modules
 - Disabled providers cannot be selected
 7. Click **Create** to finalize configuration

DRM Provider List Display

Once created, the DRM provider appears with:

Field	Information
Status	ENABLED / DISABLED
Name	Provider identifier
Provider Type	Service provider name
DRM Types	Supported technologies

Table 25: DRM Provider List Information

DRM Provider Actions

Action	Icon	Function
View	Eye	Display provider details
Disable/Enable	Toggle	Enable or disable provider
Edit	Pencil	Modify configuration
Delete	Bin	Remove provider

Table 26: DRM Provider Actions

View DRM Provider Details

Top Action Buttons:

- **Disable** - Disables provider and prevents selection by packagers
- **Edit** - Modify configuration details
- **Back** - Return to DRM Provider list

Basic Information:

- **Status** - ENABLED / DISABLED
- **Name** - DRM provider configuration name
- **Provider Type** - Service provider (e.g., Mobiotics)
- **Description** - Optional description

Supported DRM Types:

1. **FairPlay (HLS)** - Apple device protection
2. **Widevine (DASH)** - Android and web protection

Configuration Details:

Parameter	Description
Username	Authentication username for DRM service
Password	Masked password for account security
Key Server URL	Endpoint for requesting DRM keys
Tenant ID	Account/tenant identifier
Created At	Configuration creation timestamp
Updated At	Last modification timestamp

Table 27: DRM Provider Configuration Details

Stream Parameters:

Required when enabling DRM on packagers:

DRM Type	Required Parameters
FairPlay (HLS)	Content ID, Key ID (auto-generated)
Widevine (DASH)	Content ID, Key ID (auto-generated)

Table 28: DRM Stream Parameters

Edit DRM Provider

The Edit page allows modification of DRM provider configuration:

Basic Information:

- **Name** - Unique identifier for DRM provider
- **Provider Type** - DRM service provider selection
- **Description** - Optional description for reference

- **Enabled** - Toggle to enable/disable provider

Provider Configuration:

- **Username** - DRM provider account username
- **Password** - DRM provider account password
- **Key Server URL** - DRM key server endpoint for key generation
- **Tenant ID** - Tenant identifier for account management

Applying Changes:

1. Review all modifications
2. Click **Save** to apply changes
3. Click **Cancel** to discard without saving
4. Verify packagers are updated if credentials changed

Enabled / Disabled

The toggle action restarts the DRM from the platform.

Deleting Workflow:

1. Navigate to the DRM providers and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the DRM from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the DRM providers from the platform.

Deleting Workflow:

1. Navigate to the DRM providers and click the bin (Delete) icon
2. A confirmation message will appear
3. Click **Ok** to remove the DRM provider from the platform
4. Or click **Cancel** to discard the action

Origin Server Management

The Origin Server stores and serves packaged media segments produced by AES, DASH, and HLS packagers. Origins host all streaming content and serve it to CDNs and direct client requests.

Origin Server Overview

- Stores packaged media segments (HLS .ts, DASH .mp4, AES segments)

- Serves DASH .mpd manifests and HLS .m3u8 playlists
- Provides content to CDN edge servers and direct clients
- Manages media directory storage and organization
- Hosts playback URLs for client consumption
- Critical component of content delivery pipeline
- Typically deployed on same server as packagers

Steps to Add a New Origin

1. Navigate to **Origins** from left Sidebar
2. Click **Add Origin** button (top-right)
3. **Select Server** (choose server for origin deployment):
 - Typically same server hosting packagers
 - Ensure sufficient disk space for segment storage
 - Verify network connectivity to packagers
4. **Select Server Interface** (choose origin interface):
 - Usually primary server interface
 - Verify interface has internet connectivity
5. **Choose Deployment Version** (optional):
 - Select specific version if required
 - Use default version if unsure
6. **Set Media Directory** (enter storage path):
 - Standard path: /opt/es/media/packager
 - This is where packagers write output segments
 - Ensure directory exists and is writable
 - Verify sufficient disk space available
7. **Enable Origin** (toggle **Enable** = **ON**):
 - Activates origin for content serving
8. Click **Save** to finalize origin creation

Origin List Display

Once created, the Origin appears with:

Field	Information
Enabled	ENABLED / DISABLED
Live Status	RUNNING / STOPPED

Server	Hosting server name
Network Interface	Network interface name or IP address
Media Directory	Storage path
Version	Deployment version

Table 29: Origin Server List Information

Origin Actions

Action	Icon	Function
View	Eye	Display origin details
Disable/Enable	Toggle	Start or stop origin service
Edit	Pencil	Modify configuration
Delete	Bin	Remove origin

Table 30: Origin Server Actions

View Origin Details

Top Action Buttons:

- **Disable / Enable** - Temporarily stop or start origin service
- **Edit** - Modify origin configuration
- **Back** - Return to Origin list

Fields Description:

Field	Description / Example
Server	Server hosting the origin (e.g., origin-server-01)
Server Interface	Network interface in use (e.g., eth0)
Live Status	Runtime state (e.g., RUNNING)
Deployment Version	Origin service version (e.g., v2.0)
Media Directory	Segment storage path (e.g., /opt/es/media/packager)
Status	Configuration state (ENABLED / DISABLED)
Created At	Creation timestamp
Updated At	Last modification timestamp

Table 31: Origin Configuration Details

Media Directory Usage:

- Stores all packaged media output from AES, DASH, HLS packagers
- Directory structure organized by content and stream
- Packagers write segments to this location
- Origin serves content from this directory

Edit Origin

The Edit page allows modification of deployment and storage settings:

Configuration Fields:

- **Server** - Server hosting the origin
 - Typically the same server hosting packagers
 - Changing requires proper server preparation
- **Server Interface** - Primary server interface for connectivity
- **Deployment Version** - Origin service version selection
- **Media Directory** - Absolute path for segment storage
 - Example: /opt/es/media/packager
 - Must exist and be writable by packagers
 - Requires sufficient disk space
- **Enable Origin** - Activation toggle
 - Enable - Activate origin immediately
 - Disable - Keep configured but inactive

Applying Changes:

1. Review all modifications
2. Click **Save** to apply changes
3. Click **Cancel** to discard without saving
4. Monitor origin status after updates

Enabled / Disabled

The toggle action restarts the origin from the platform.

Deleting Workflow:

1. Navigate to the Origins and click the toggle (enabled/disabled) icon
2. A confirmation message will appear
3. Click **Ok** to enabled/disabled the origin from the platform
4. Or click **Cancel** to discard the action

Delete

The Delete action removes the Origin from the platform.

Deleting Workflow:

5. Navigate to the Origins and click the bin (Delete) icon
 6. A confirmation message will appear
 7. Click **Ok** to remove the Origin from the platform
 8. Or click **Cancel** to discard the action
-

Deployment & Best Practices

General Deployment Guidelines

- Ensure all input streams are properly configured before deployment
- Verify transcoder and packager associations and dependencies
- Validate DRM settings if encryption is required
- Test connectivity to all origin servers before production
- Monitor deployment status during initial rollout
- Maintain records of deployment versions and dates
- Plan capacity based on content and audience size
- Implement redundancy for critical components
- Set up monitoring and alerting for all components

Pre-Deployment Checklist

Verification Item	Status
Input Streams Configured	☐
Transcoders Assigned	☐
Packagers Created (AES/DASH/HLS)	☐
ABR Profiles Validated	☐
DRM Settings Verified	☐
Origin Servers Ready	☐
CDN Integration Complete	☐
Network Connectivity Tested	☐

Monitoring Configured	☐
Failover Strategy Defined	☐

Table 32: Pre-Deployment Verification Checklist

Input Stream Management Best Practices

- Use descriptive, identifiable channel names for easy management
- Maintain unique Content IDs across all channels for data integrity
- Verify source URL availability before registration
- Test input connectivity in staging environment first
- Monitor input stream health continuously
- Document all critical source information
- Establish fallback input streams for redundancy
- Maintain input source contact information
- Review and update input addresses regularly
- Archive historical configuration changes

Transcoder Configuration Best Practices

- Select appropriate ABR profiles for target audience demographics
- Configure renditions based on viewer network capabilities
- Test output quality before production deployment
- Monitor transcoder resource utilization regularly
- Balance quality and bandwidth based on service requirements
- Implement server load balancing for scalability
- Use consistent naming conventions for transcoders
- Document server capacity and performance metrics
- Schedule maintenance during low-traffic periods
- Track and optimize transcoding efficiency

Packager Optimization Best Practices

- Configure appropriate segment and buffer durations for target devices
- Enable DRM protection for premium and copyrighted content
- Test playback across all target devices before production
- Monitor packager performance metrics continuously
- Regularly rotate DRM keys for security compliance

- Implement geographic distribution for packagers
- Use CDN origin shields to reduce packager load
- Monitor manifest generation and segment delivery times
- Implement segment caching strategies
- Validate multi-bitrate switching in player tests

System Monitoring Best Practices

- Review dashboard status regularly (ideally continuously)
 - Set up alerts for failed components and critical events
 - Maintain service uptime targets and SLA compliance
 - Document all configuration changes for audit trails
 - Conduct regular system audits and performance reviews
 - Monitor bandwidth utilization and CDN costs
 - Track player errors and quality metrics
 - Implement logging for troubleshooting and analysis
 - Use metrics to forecast capacity needs
 - Conduct regular disaster recovery testing
-

Support & Troubleshooting

Common Issues and Resolution Steps

Issue	Symptoms	Resolution Steps
Input Stream Not Connecting	No signal, dashboard shows red	Verify source URL, check network connectivity, test UDP/RTMP routing
Transcoder Failing	Stopped status, no output	Check server resources (CPU/memory), verify input source active, review logs
Missing Packager Output	No segments created	Verify transcoder association, confirm renditions configured, check disk space
DRM Errors	License failures, playback blocked	Validate DRM provider credentials, verify key server accessibility, check tenant ID
Playback Issues	Buffering, quality drops	Test across multiple devices/networks, check bandwidth, verify segment delivery
High Latency	Excessive delay in playback	Reduce buffer duration, adjust segment settings, check network latency

Video Quality Degradation	Lower resolution/bitrate	Monitor transcoder CPU, reduce concurrent streams, scale infrastructure
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Table 33: Common Issues and Troubleshooting Guide

Troubleshooting Workflow

1. **Check Dashboard** - Verify component status (green/yellow/red)
2. **Review Component Details** - View specific service status and logs
3. **Test Connectivity** - Verify network paths and protocol connectivity
4. **Validate Configuration** - Confirm all parameters are correct
5. **Review System Resources** - Check CPU, memory, disk, bandwidth
6. **Check Error Logs** - Review system logs for error messages
7. **Isolate Issue** - Determine if issue is input, transcoding, packaging, or origin
8. **Apply Resolution** - Implement corrective action based on findings
9. **Monitor Recovery** - Verify issue is resolved and system is stable
10. **Document Incident** - Record issue, cause, and resolution for future reference

Support Resources

- **vLive Documentation** - Comprehensive platform guides and API references
- **Technical Support Team** - Contact support for complex issues
- **Monitoring Tools** - Use built-in dashboard for real-time visibility
- **Log Analysis** - Review component logs for detailed error information
- **Configuration Validation** - Verify all settings against best practices
- **Performance Metrics** - Use metrics to identify optimization opportunities

End of Document
